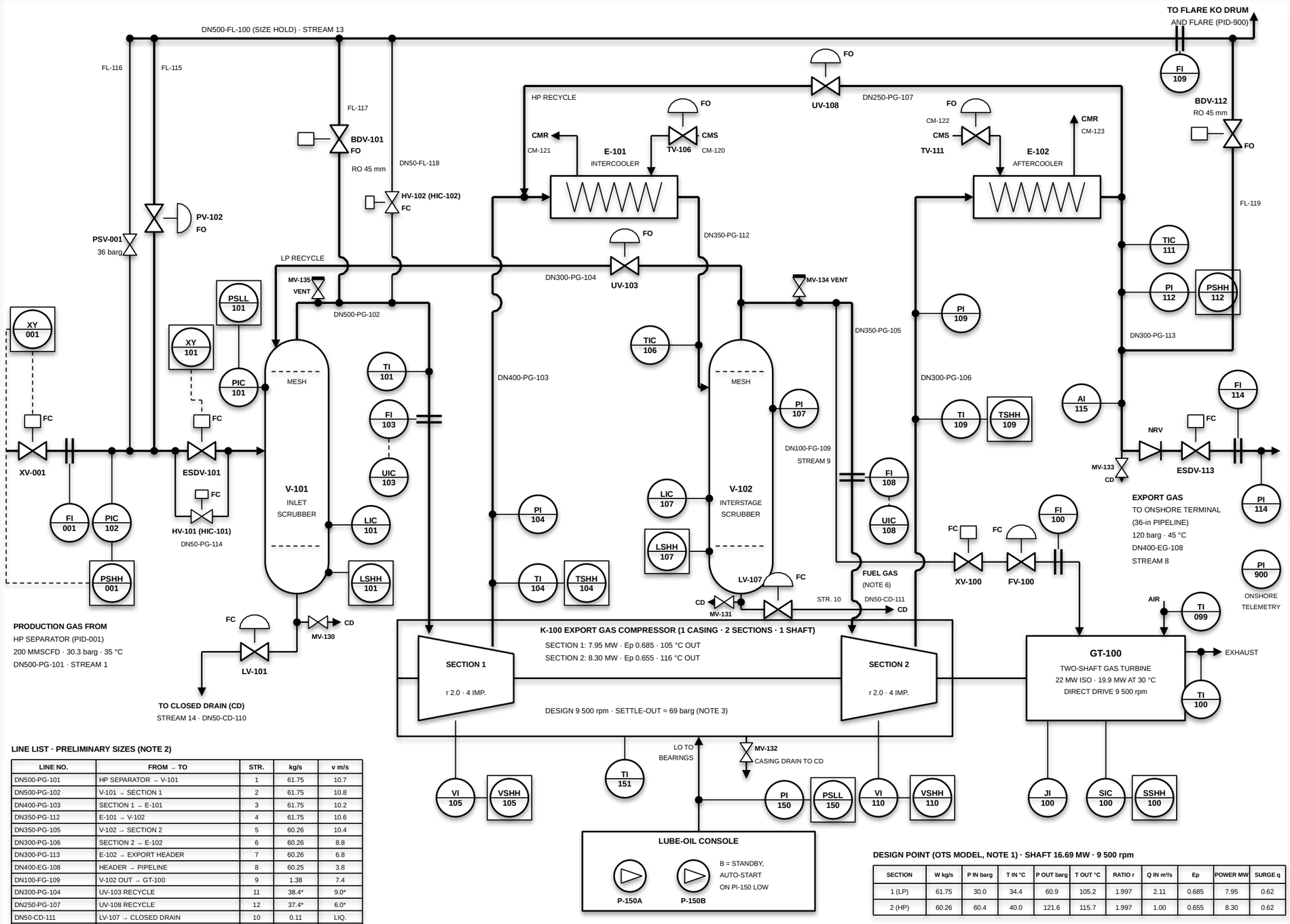


LEGEND

- CONTROL VALVE (DCS) · FC / FO = FAIL CLOSED / OPEN
- ESDV / BDV / XV ON-OFF VALVE (SIS), FAIL AS MARKED
- REMOTE HAND VALVE HV (HIC IN THE DCS)
- MANUAL VALVE MV (DRAIN / VENT / ISOLATION)
- CHECK VALVE (NRV)
- PRESSURE SAFETY VALVE (PSV) TO FLARE
- FLOW ELEMENT / RESTRICTION ORIFICE (RO)
- CENTRIFUGAL COMPRESSOR SECTION
- PUMP
- DCS INSTRUMENT (TRANSMITTER IMPLIED, SAME NUMBER)
- SIS TRIP FUNCTION (LOGIC SOLVER SEPARATE FROM DCS)
- LEADER: SIS ON A DCS BUBBLE SHARES ITS TRANSMITTER
- INSTRUMENT SIGNAL (ONLY TRIP / SOLENOID LINKS DRAWN)
- PROCESS GAS LINE
- UTILITY / DRAIN / RELIEF / FUEL LINE
- CONNECTION (TEE)
- CROSSING WITHOUT CONNECTION (VERTICAL JUMPS)

LINE NUMBER DNxxx-SVC-nnn · SERVICE CODES
PG PROCESS GAS · EG EXPORT GAS · FG FUEL GAS · FL FLARE
CD CLOSED DRAIN · CM COOLING MEDIUM · STREAM n = PFD / PROCESS TAB

INTERLOCK SCHEDULE (SIS) · FROM THE OTS C&E DATA

TAG	CAUSE	TRIP SET	DELAY	PRE-ALARM	ACTS
ESD-1	ESD PUSH-BUTTON (CCR)	—	0 s	—	I-1
ESD-2	ESD PUSH-BUTTON + BLOWDOWN	—	0 s	—	I-2
LSHH-101	V-101 LEVEL	≥ 85 %	3 s	65 %	I-1
LSHH-107	V-102 LEVEL	≥ 85 %	3 s	65 %	I-1
TSHH-104	SECTION 1 DISCH. TEMP.	≥ 150 °C	3 s	125 °C	I-1
TSHH-109	SECTION 2 DISCH. TEMP.	≥ 150 °C	3 s	130 °C	I-1
VSHH-105	SECTION 1 VIBRATION	≥ 90 µm	3 s	60 µm	I-1
VSHH-110	SECTION 2 VIBRATION	≥ 90 µm	3 s	60 µm	I-1
PSLL-150	LUBE-OIL HEADER (TURNING)	≤ 1.2 barg	2 s	1.8 barg	I-1
SSHH-100	OVERSPEED (SIC-100 SIGNAL)	≥ 110 %	0.5 s	104 %	I-1
PSHH-112	DISCHARGE HEADER	≥ 132 barg	2 s	126 barg	I-1
PSLL-101	SUCTION V-101 (RUN > 60 %)	≤ 20 barg	5 s	28.5 barg	I-1
GT-100	GT TRIP / FLAME-OUT	—	0 s	—	I-1
PSHH-001	PRODUCTION HEADER	≥ 34.5 barg	3 s	32.5 barg	XV-001

I-1 COMPRESSOR TRIP (PSD): XV-100 FUEL SHUT (GT COASTS DOWN), ESDV-101 + ESDV-113 SHUT, UV-103 / UV-108 100 % OPEN (UIC TO MAN), PIC-101 + SIC-100 TO MAN (SIC OUT 0 %), PIC-102 OUTPUT ≥ 80 % (PRODUCTION TO FLARE). LATCHED, FIRST-OUT ON THE SIS TAB.
I-2 = I-1 + BDV-101 / BDV-112 OPEN (RO 45 mm) + XV-001 SHUT (WELLS SHUT IN).
XV-001: PSHH-001 SHUTS THE WELLS ONLY (NO I-1); REOPEN BELOW 33.5 barg.
TRIPS ACT IN RUN / START; RESET REFUSED WHILE A CAUSE HOLDS OR SPEED > 5 %.

LINE LIST · PRELIMINARY SIZES (NOTE 2)

LINE NO.	FROM → TO	STR.	kg/s	v m/s
DN500-PG-101	HP SEPARATOR → V-101	1	61.75	10.7
DN500-PG-102	V-101 → SECTION 1	2	61.75	10.8
DN400-PG-103	SECTION 1 → E-101	3	61.75	10.2
DN350-PG-112	E-101 → V-102	4	61.75	10.6
DN350-PG-105	V-102 → SECTION 2	5	60.26	10.4
DN300-PG-106	SECTION 2 → E-102	6	60.26	8.8
DN300-PG-113	E-102 → EXPORT HEADER	7	60.26	6.8
DN400-EG-108	HEADER → PIPELINE	8	60.25	3.8
DN100-FG-109	V-102 OUT → GT-100	9	1.38	7.4
DN300-PG-104	UV-103 RECYCLE	11	38.4*	9.0*
DN250-PG-107	UV-108 RECYCLE	12	37.4*	6.0*
DN50-CD-111	LV-107 → CLOSED DRAIN	10	0.11	LIQ.
DN50-CD-110	LV-101 → CLOSED DRAIN	14	0.22	LIQ.

* AT THE SURGE LINE q = 0.62, DESIGN SPEED (FULL RECYCLE CASE)

DESIGN POINT (OTS MODEL, NOTE 1) · SHAFT 16.69 MW · 9 500 rpm

SECTION	W kg/s	P IN barg	T IN °C	P OUT barg	T OUT °C	RATIO r	Q IN m³/s	Ep	POWER MW	SURGE q
1 (LP)	61.75	30.0	34.4	60.9	105.2	1.997	2.11	0.685	7.95	0.62
2 (HP)	60.26	60.4	40.0	121.6	115.7	1.997	1.00	0.655	8.30	0.62

GENERAL NOTES

1. DESIGN BASIS (ASSUMED FOR TRAINING, TBC): 200 MMSCFD (61.75 kg/s) GULF OF THAILAND GAS, MW 22.3 (CO2 12 %), γ 1.258; 30 → 120 barg; N = 2 SECTIONS, r = 3.96^(1/2) = 1.99 EACH (ONE STAGE: T2 ≈ 192 °C > SEAL CEILING 150 °C). Ep FROM LECTURE CHART; T2 = T1·r^((γ−1)/(γ·Ep)). NUMBERS = OTS DESIGN POINT. MASS BALANCE 61.75 IN ≈ 60.25 EXPORT + 1.38 FUEL + 0.11 CONDENSATE (+0.005 ACCUMULATION) kg/s; V-101 LIQUIDS 0.22 kg/s ARRIVE WITH THE GAS.
2. LINE SIZES PRELIMINARY: VELOCITY AT DN BORE, CRITERIA v ≤ 20 m/s AND ρv² ≤ 15 000 Pa (ASSUMED); ALL ≤ 10.8 m/s AND ≤ 7 500 Pa. FLARE LINES: HOLD (FLARE STUDY).
3. SETTLE-OUT AFTER A TRIP ≈ 69 barg (MODEL): SUCTION SIDE (V-101, SECTION 1) DESIGN PRESSURE 80 barg (TBC). PSV-107 80 barg ON V-102 AND PSV-112 138 barg ON THE DISCHARGE: HOLD (NOT DRAWN). BLOWDOWN COOLS BY EXPANSION: FLARE AND BLOWDOWN LINES LTCS, MDMT −46 °C (TBC).
4. UIC-103/108 SURGE MARGIN = (q/q_SURGE − 1)·100 % FROM FT, PT, TT; CONTROL LINE 10 %; FAST OPEN BELOW 5 %. UV-103/108 ALSO HAVE ZI-103/108 (POSITION).
5. CASCADE: PIC-101 (SUCTION) → SIC-100 SPEED SP (0–100 % = 70–105 %) → FV-100 FUEL. DCS SIGNAL LINES NOT DRAWN; EACH DCS BUBBLE HAS ITS OWN TRANSMITTER (PT/TT/LT/FT/VT, SAME NUMBER, e.g. TT-106 FOR TIC-106). PSHH-001 SHARES PT-001 (PIC-102), PSLL-101 PT-101 (PIC-101), SSHH-100 THE SPEED PICK-UP.
6. FUEL GAS FROM V-102 OUTLET VIA FUEL-GAS SKID (PCV TO 25 barg, HEATER, FILTER NOT SHOWN). LUBE-OIL CONSOLE, SEAL-GAS PANEL, CM SYSTEM (P-160, TI-160): VENDOR / SEPARATE SHEETS.
7. DRAINS MV-130…133 TO CLOSED DRAIN, VENTS MV-134/135 BLINDED; ALL OTHER HIGH-POINT VENTS / LOW-POINT DRAINS DN25 (TYP., NOT SHOWN). INSTRUMENTS C/W ROOT & TEST VALVES.

OPERATING NOTES AND OPEN ITEMS

A. START-UP: PRESSURISE WITH HV-101 UNTIL ΔP ACROSS ESDV-101 ≤ 3 bar, OPEN ESDV-101, START GT-100 ON FULL RECYCLE. GT PERMISSIVES: SIS READY, PI-150 ≥ 2.0 barg, P-160 RUNNING, PI-101 ≥ 20 barg, UV-103/108 ≥ 90 %, LEVELS < 60 %, BDVs SHUT.
B. AFTER A PSD THE LOOP SETTLES OUT: VENT WITH HV-102 TO HEADER PRESSURE BEFORE REOPENING ESDV-101.
C. ANTI-SURGE, FLARE AND BLOWDOWN VALVES ARE CHOKED WHEN P2/P1 < 0.553.
D. P-150B STANDBY LUBE-OIL PUMP AUTO-STARTS AT PI-150 1.8 barg (BPCS).
E. HOLD / TBC: PSV-107, PSV-112, FLARE LINE SIZES, HV-101/102 FAIL POSITION (FC ASSUMED), DESIGN PRESSURES, CM SIDE (E-101 7.3 MW, E-102 7.7 MW).
F. REV B = ALTERNATIVE VERSION (EPC STANDARD). REV A STAYS THE SCREEN REFERENCE USED BY THE DCS, THE DECKS AND THE EXERCISES; SAME TAGS.

REV	DESCRIPTION	DATE
A	SCREEN REFERENCE USED BY THE DCS, DECKS AND EXERCISES	25-SEP-26
B	ALTERNATIVE VERSION, EPC STANDARD	12-OCT-26

CHEM ENG CHULALONGKORN UNIVERSITY · OTS

P&ID — GAS EXPORT COMPRESSOR K-100
ALTERNATIVE VERSION

DRAWN: Apinan Soottitantawat	CHK'D: --	APP'D: --	DATE: 12-OCT-26
DWG NO. KC-PID-100	REV B	SCALE N.T.S.	SHEET 1 OF 1